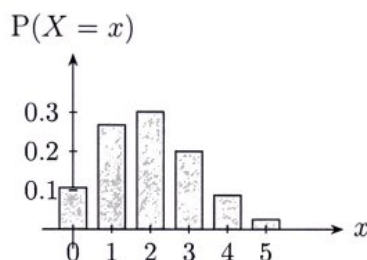


4.19 Quiz: Normal and Binomial Distribution

1. A market stall sells bags of apples. It is known that 20% of the apples have bruises. A bag contains 10 randomly selected apples. Let X be the number of bruised apples in the bag.



- (a) Find the expected number of bruised apples in a bag, $E(X)$. [1]
 (b) Find the probability that none of the apples are bruised. [2]
 (c) Find $P(X = 3)$. [2]
 (d) Find $P(X \leq 3)$. [2]

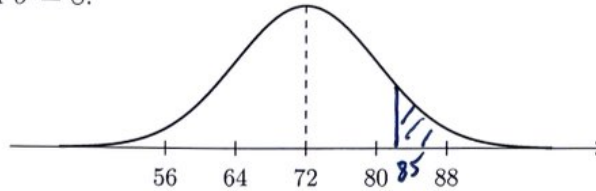
a) $10 \cdot 0.20 = 2$

b) $P(X=0) = 0.80^{10} = 0.107371..$
 ≈ 0.107

c) $\text{bin PDF}(10, 0.2, 3) = 0.201327..$
 ≈ 0.201

d) $\text{binom CDF}(10, 0.2, 0, 3) = 0.8791..$
 ≈ 0.879

2. The marks on a mathematics test follow a normal distribution with mean $\mu = 72$ and standard deviation $\sigma = 8$.



- (a) Find the probability that a randomly selected student scores higher than 85. [2]
 (b) Find the probability that a student's mark is between 60 and 80. [2]
 (c) 75% of the students score less than h marks. Find h . [3]

$$\begin{aligned} \text{a) } \text{Norm CDF}(85, \infty, 72, 8) &= 0.05208... \\ &\approx 0.0521 \\ \text{b) } \text{Norm CDF}(60, 80, 72, 8) &= 0.77453... \\ &\approx 0.775 \\ \text{c) } \text{Inv Norm}(0.75, 72, 8) &= 77.3959... \\ &\approx 77.4 \end{aligned}$$

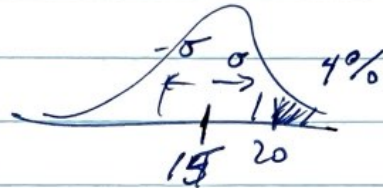
Answer this section on a separate piece of lined paper

3. The waiting time T (minutes) at a bus stop follows a normal distribution with mean 15 minutes and standard deviation σ minutes. It is known that 4% of waits are longer than 20 minutes.
- (a) Find the value of σ . [3]
 (b) Find $P(T < 12)$. [2]
 (c) On a particular day, 10 people wait at this bus stop. Find the probability that at most 2 of them wait more than 20 minutes. [3]
 (d) A person is selected at random. Given that they waited less than 15 minutes, find the probability that they waited less than 10 minutes. [3]

4.19 a Normal & Binomial Distributions
25 March 2026 (13 March)

Solutions

3. a) $\text{InvNormal}(0.96, 0, 1)$
 $= 1.75069...$



$$z = \frac{20 - 15}{\sigma} = 1.75069...$$

$$\sigma = 2.85602...$$
$$\approx 2.86$$

b) $\text{normCDF}(-\infty, 12, 15, 2.85602...)$
 $= 0.146764...$
 ≈ 0.147

c) $\text{binomCDF}(x \leq 2, 10 \text{ trials}, 0.04)$
 $10, 0.04$
 $= 0.993786...$
 ≈ 0.994

d) $P(T < 15) = 0.5$
 $P(T < 10) = \text{normCDF}(-\infty, 10, 15, 2.85602...)$
 $= 0.04$
 $P(T < 10 | T < 15) = \frac{0.04}{0.5} = 0.08$